higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

FLUID MECHANICS N6

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This marking guideline consists of 8 pages.

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QUESTION 1

1.1 Mass per unit volume

OR

Mass divided by volume
(**NOTE**: If the formula is used as a definition, it must be marked wrong.) (1)

1.2 The sum of energies is constant provided there is NO energy leaving or entering the system or the system is closed. (2)

1.3 1.3.1 They have the same friction factors✓ as they are both new/the same pipes.✓

1.3.2 Their hydraulic mean depths are different✓ as they have different diameters.✓

1.3.3 They have the same friction heads✓ as they are connected in parallel.✓

1.3.4 The sum of their discharges is equal to the total from the common pump✓ as they are connected in parallel.✓

(4 × 2) (8)

1.4 1.4.1 ρ = 0,78 × 1 000✓
= 780 kg/m^3✓ (2)

1.4.2 $A = \frac{Q}{V}$

$= \frac{0{,}02417}{9}$✓

$\frac{\pi d^2}{4}$ = 0,002685 m^2✓

d = 0,0585 m

= 58,47 mm✓ (3)

1.4.3 $f = 0{,}005\left(1 + \frac{1}{40\,d}\right)$

$= 0{,}005\left(1 + \frac{1}{40\ x\ 0{,}0585}\right)$✓

= 0,00714✓

$h_f = \frac{4\,fl\,v^2}{2\,gd}$ alternatively $h_f = \frac{fl\,Q^2}{3\,d^5}$

$= \frac{4\ x\ 0{,}0071\ x\ 4\ 000\ x\ 9^2}{2\ x\ 9{,}81\ x\ 0{,}0585}$ ✓ =

(4)

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$$\frac{0,0071\ x\ 4\ 000\ x\ 0,02417^2}{3\ x\ 0,0585^5}$$ ✓

= 8 063,59 m ✓ = 8 132,535 m✓

1.5 1.5.1 A tapered pipeline has a different cross sectional area/diameter at its entry and at its exit. (1)

1.5.2 Q = VA

$= 1,105 \times \frac{\pi(1,026)^2}{4}$ ✓

= 0,914 m^3/s

= 913,581 ℓ/s✓ (2)

1.5.3 $h_L = \frac{0,5v^2}{2g}$

$= \frac{0,5\ x\ (1,105)^2}{2\ x\ 9,81}$ ✓

= 0,0311 m✓ (2)

1.5.4 $h_L = \frac{(v_1 - v_2)^2}{2g}$

$= \frac{(1,105 - 0)^2}{2\ x\ 9,81}$ ✓

= 0,0622 m✓ (2)

1.5.5 $h_f = \frac{4flv^2}{2gd}$ Alternatively $h_f = \frac{flQ^2}{3d^5}$

$= \frac{4\ x\ 0,03\ x\ 1\ 100\ x\ 1,105^2}{2\ x\ 9,81\ x\ 1,026}$ ✓ =

$\frac{0,03\ x\ 1\ 100\ x\ 0,868^2}{3\ x\ 1,026^5}$ ✓

= 8,007 m✓ = 8,083 m✓ (2)

[29]

QUESTION 2

2.1

$A = \frac{1}{2}\pi r^2$

$= \frac{1}{2} \times \pi(1,5)^2$ ✓

$= 3,534\ m^2$ ✓

Alternatively

$A = \frac{1}{8}\pi d^2$

$= \frac{1}{8}\pi(3)^2$ ✓

$= 3,53\ m^2$ ✓

$P = \pi r$

$= \pi \times 1,5$ ✓

$= 4,712$ m

Alternatively

$P = \frac{1}{2}\pi d$

$= \frac{1}{2}\pi \times 3$ ✓

$= 4,71$ m ✓

$m = \frac{A}{P}$

$= \frac{3,53}{4,71}$ ✓

$= 0,75$ m ✓

$i = \frac{1}{1850}$

$= 0,541 \times 10^{-3}$

$Q = AC\sqrt{mi}$

$= 3,53 \times 60 \times \sqrt{0,75\ x\ 0,54\ x\ 10^{-3}}$ ✓

$= 4,27\ m^3/s$ ✓

$= 4\ 269,71\ \ell/s$ ✓ (9)

2.2

$Q = Cd \times \frac{8}{15} \times \sqrt{2g} \tan\frac{\theta}{2} \times H^{2,5}$

$H = \sqrt[2,5]{\frac{Q\ x\ 15}{C_d\ x\ 8\ x\ \sqrt{2g}\ x \tan\frac{\theta}{2}}}$ ✓✓

$= \sqrt[2,5]{\frac{0,94\ x\ 15}{0,65\ x\ 8\ x\ \sqrt{2\ x\ 9,81}\ x \tan\frac{90}{2}}}$ ✓

$= 0,822$ m

$= 822$ mm ✓ (Award ½ mark if the answer is left in m.) (4)

2.3 2.3.1

$V_A = \sqrt{\frac{g x^2}{2y}}$

$= \sqrt{\frac{9,81\ x\ 3,2^2}{2(1)}}$ ✓ (4)

= 7,087 m/s ✓

$t = \frac{S}{V}$

$= \frac{3,2}{7,087}$ ✓

= 0,452 s ✓

2.3.2 $V_A = C_V\sqrt{2gh}$

$7,087 = 0,97\sqrt{2\ x\ 9,81\ x\ h}$ ✓

h = 2,721 m ✓ (2)

2.3.3 $Q_{th} = V_{th} \times A_{th}$

$= \sqrt{2gh} \times \frac{\pi d^2}{4}$

$= \sqrt{2\ x\ 9,81\ x\ 2,72}$ ✓ × $\frac{\pi(0,05)^2}{4}$ ✓

= 14,346 ℓ/s ✓ (3)
[22]

QUESTION 3

3.1
- Propeller fans
- Screw fans (2)

3.2

$a = b^2$	$V = \frac{Q}{a}$	$S = l\,b^2$
$= (0,75)^2$	$= \frac{9,2}{0,5625}$	$= 50 \times 0,75^2$
= 0,5625 m²✓	= 16,36 m/s✓	= 28,125 m²✓

$Pr = \frac{kSV^2}{a}$

$= \frac{0,003\ x\ 28,125\ x\ (16,3556)^2}{0,5625}$ ✓

= 40,13 Pa✓ (5)

3.3 $Pr = \rho g H_{at}$

$$H_{at} = \frac{130\ x\ 10^3}{1,1\ x\ 10^3 x\ 9,81}$$

$= 12,047$ m✓

$Pr = \rho g H_{sep}$

$$H_{sep} = \frac{1,9\ x\ 10^3}{1,1\ x\ 10^3 x\ 9,81}$$

$= 0,176$ m✓

$Ha = Hat - hs - hsep$

$= 12,047 - 5 - 0,176$✓

$= 6,871$ m✓

$$h_{as} = \frac{l}{g} \times \frac{D^2}{d^2} \times \omega^2 R$$

$$6,871 = \frac{10}{9,81} \times \frac{0,09^2}{0,13^2} \times \omega^2\ x\ \frac{0,26}{2}$$ ✓

$\omega = 10,401$ rad/sec✓

$$N = \frac{10,401\ \ x\ 60}{2\pi}$$ ✓

$= 99,322$ r/min✓ (8)

3.4 It is the formation of a vacuum in a cylinder or a suction pipe.✓

CAUSES OF CAVITATION
- Plunger speed too high
- Pressure too high in the cylinder
- Suction pipe diameter too small (Any 2 × 1) (3)

3.5 3.5.1 $h = \frac{P}{\rho g}$

$= \frac{285\ x\ 10^3}{10^3\ x\ 9,81}$ ✓

= 29,052 m✓

3.5.2 $U_o = \frac{\pi\ D\ N}{60}$

$= \frac{\pi\ x\ 0,38\ x\ 1430}{60}$ ✓

= 28,452 m/s✓

3.5.3

$V_{fo} = \frac{Q}{\pi\ x\ D\ x\ Width}$

$= \frac{0,0233}{\pi\ x\ 0,38\ x\ 0,012}$ ✓

= 1,626 m/s✓

3.5.4 $V_{wo} = \frac{gH}{U_o\ x\ \eta_m}$

$= \frac{9,81\ x\ 29,05}{28,45\ x\ 0,74}$ ✓

= 13,536 m/s✓

3.5.5 $X = U_o - V_{wo}$

= 28,452 - 13,536✓

= 14,916 m/s✓

3.5.6 $\phi = \tan^{-1}\left(\frac{V_{fo}}{X}\right)$

$= \tan^{-1}\left(\frac{1,63}{14,92}\right)$ ✓

$= 6,23^0$✓

(6 × 2) (12)
[30]

QUESTION 4

4.1 4.1.1

$$V_1 = \frac{0,32 \ x \ 4}{\pi \ x \ (0,35)^2} = 3,326 \text{ m/s}\checkmark$$

$$V_2 = \frac{0,32 \ x \ 4}{\pi \ x \ (0,65)^2} = 0,964 \text{ m/s}\checkmark$$

$$\frac{Pr_1}{\rho g} + \frac{v^2}{2g} + z_1 - H_e = \frac{Pr_2}{\rho g} + \frac{v^2}{2g} + z_2\checkmark$$

$$\frac{150}{9,81} + \frac{3,326^2}{2 \ x \ 9,81} + 2 - H_e = \frac{-35}{9,81} + \frac{0,964^2}{2 \ x \ 9,81} + 0\checkmark$$

$$H_e = 21,375 \text{ m}\checkmark$$

(7)

4.1.2

$$P = \rho g Q \ H_e$$
$$= 1 \ 000 \ x \ 9,81 \times 0,32 \times 21,375\checkmark$$
$$= 67,099 \text{ kW}\checkmark$$

(2)

4.2 4.2.1

$$V_A = C_V\sqrt{2gh}$$
$$= \sqrt{2 \ x \ 9,81 \ x \ 250}\ \checkmark$$
$$= 70,036 \text{ m/s}\ \checkmark$$

For maximum efficiency $U = 0,5 \ V_A$

$U = 0,5 \ (70,036)\checkmark$

$= 35,018 \text{ m/s}\checkmark$

$$U = \frac{\pi \ D \ N}{60}$$

$$35,018 = \frac{\pi \ x \ 1,1 \ x \ N}{60}\checkmark$$

$= 607,992 \text{ r/min}\checkmark$

(6)

4.2.2

$P = \rho QU \ (V - U) [1 + \cos (180^\circ - y)]$

$=10^3 \times 0,0717 \times 35,018 \times (70,036 - 35,018)[1 + \cos (180^\circ - 163^\circ)]\checkmark$

$=172,003 \text{ kW}\checkmark$

(2)

4.2.3

$$\eta = \frac{U}{gh} \ (V - U) \ (1 + \cos (180^\circ - y) \times 100\%$$

$$= \frac{35,018}{9,81 \ x \ 250} \ (70,036 - 35,018) \ [1 + \cos (180^\circ - 163^\circ)] \times 100\%\checkmark$$

$= 97,815\%\checkmark$

(2)

[19]

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TOTAL: 100

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